(1)

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I AB. Sai Saranya (1921260020) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: AB Saranya

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	2	2
Concept Mapping Structure	25%	3	2	2
Linking & Relationships	20%	2	1	2
Application & Analysis	20%	2	2	2
Explanation & Clarity	15%	1	1	1

Total Marks 27 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:
DataLink→MACAddress→LocalDelivery
Network → IP Address → Global Delivery
Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:
ErrorControl
→DataLinkLayer→ErrorDetection
→ Transport Layer → Error Correction
Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:
Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖
Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

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FACULTY - R. SHANKAR

Course : Computer Networks
Course Code: CSA0709

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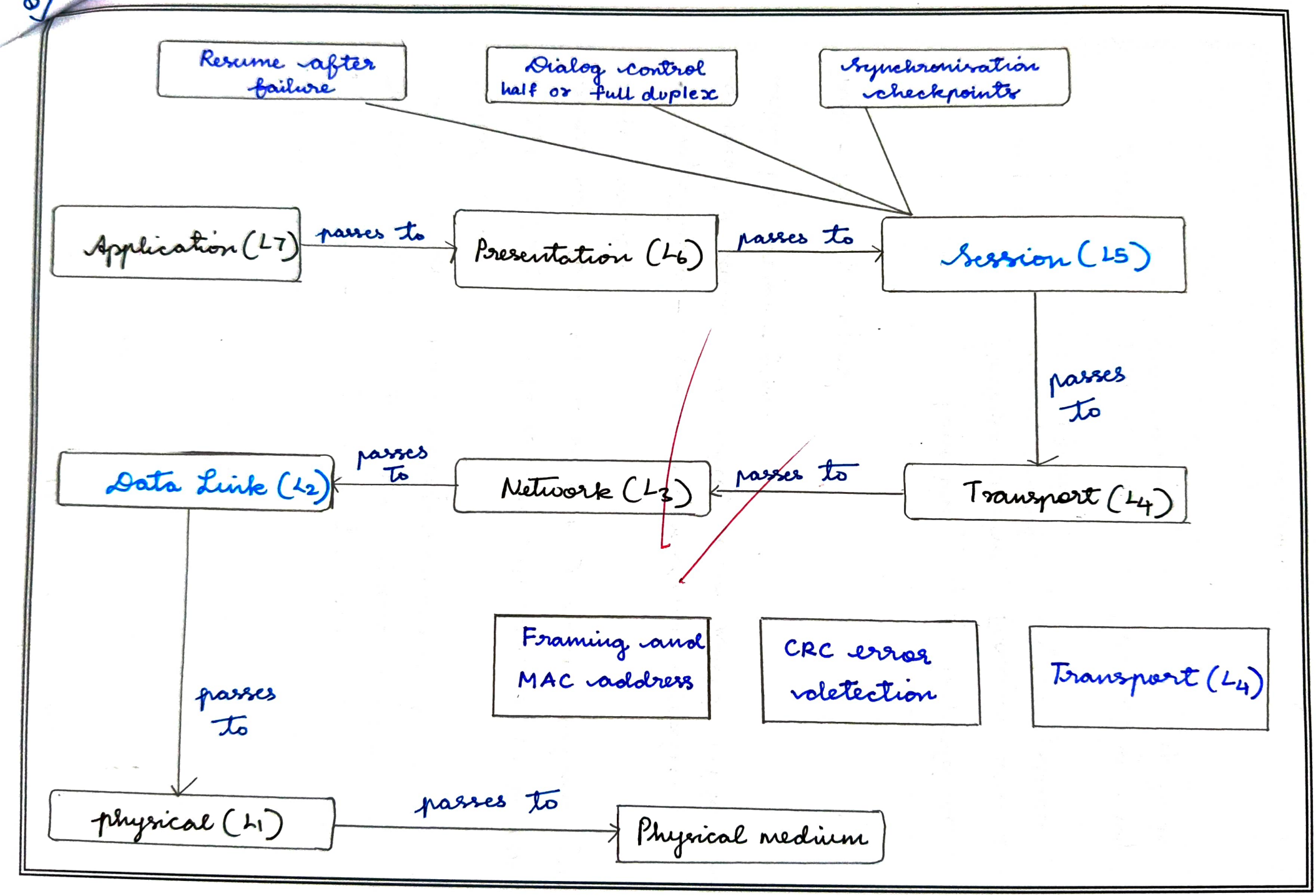
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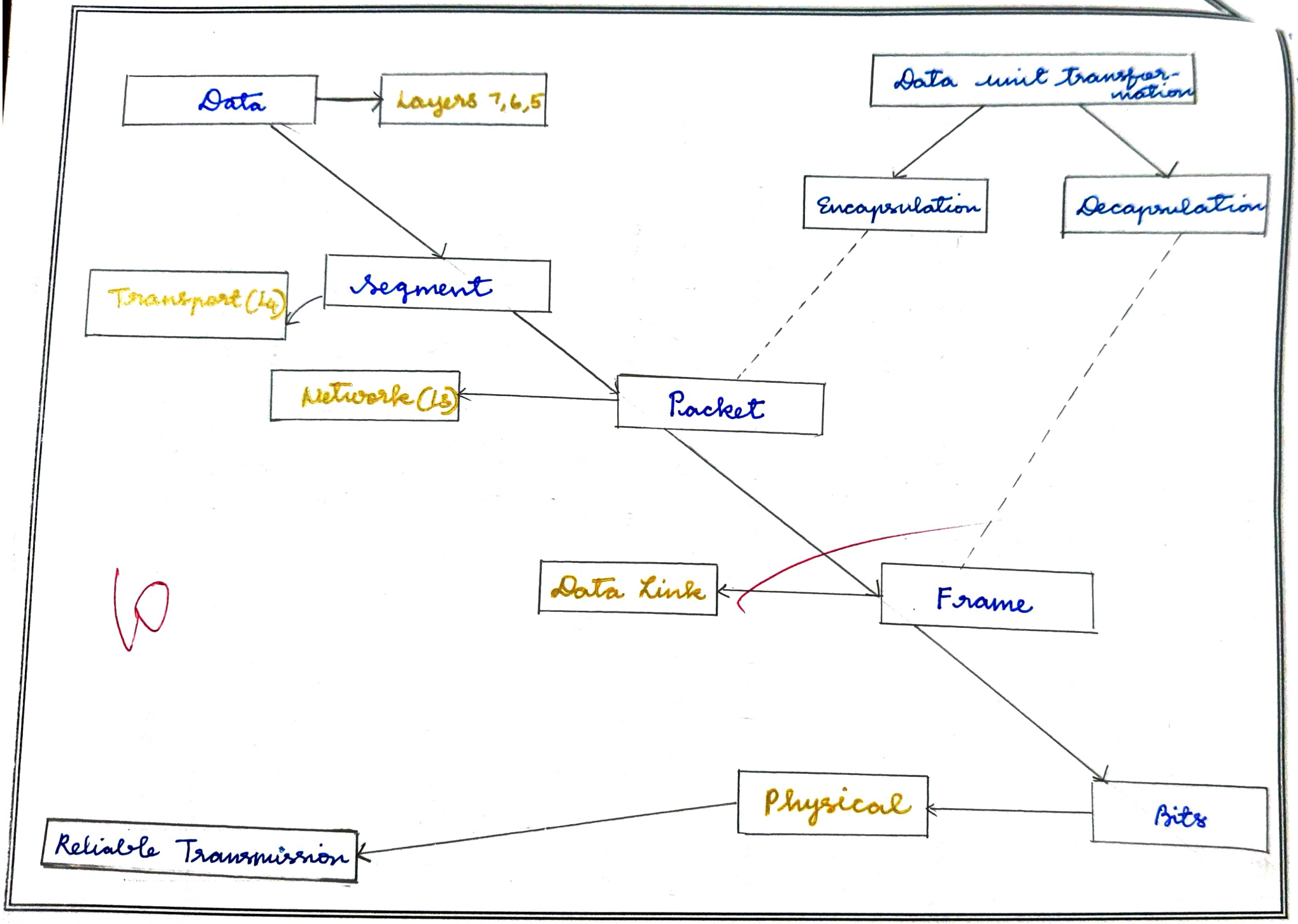
1. Missing layers

- Session (L5) - Sets up, manages and ends sessions. It gives dialog control and synchronisation check-points, so a broken transfer resumes instead of restarting.
- Data Link (L2) - Packs bits into frames with MAC addresses. It detects errors (CRC) and controls flow and access.
- Importance - Session keeps communication organised and recoverable, and Data link makes each hop error-free.

Table - Missing Layers

Layer	Function	Importance
Session (L5)	Manages sessions, dialog control, checkpoints	Transfer resumes after failure
Data Link (L2)	Frames, MAC address, CRC error detection	Error-free - hop - to hop link


Resume after failure
Dialog control half or full duplex
Synchronization checkpoints
Application (L7)
passes to
Presentation (L6)
passes to
Session (L5)
passes to
Transport (L4)
passes to
Network (L3)
passes to
Data Link (L2)
passes to
physical (L1)
passes to
Physical medium
Framing and MAC address
CRC error detection
Transport (L4)


Data
Layers 7,6,5
Data unit transfor-mation
Encapsulation
Decapsulation
Segment
Transport (L4)
Network (L3)
Packet
Data Link
Frame
Physical
Bits
Reliable Transmission

(Qn 2)

Layer & Data Unit Mapping

- Transport Layer → Segment :

Chunks raw data and adds port numbers and sequence numbers for process-to process delivery

- Network layer → Packet :

Encapsulates the packet and adds MAC addresses for node-node local delivery plus an FCS trailer for error detection.

Analysis – How Transformation Supports Transmission

- Tiered Addressing – Combines port numbers (application), IP address, MAC addresses to ensure precise addressing.

- Error Control & Integrity – The Frame trailer (CRC/FCS) allows the receiving node to discard corrupted bits immediately.

- Decapsulation – At the destination, each layer strips its corresponding header progressively upward until the original application message is successfully reconstructed.

(Data link)

Resume after failure in process

Network

MAC address

IP address

Local delivery

Global delivery

Complete data delivery

IP address: PC to server, unchanged

